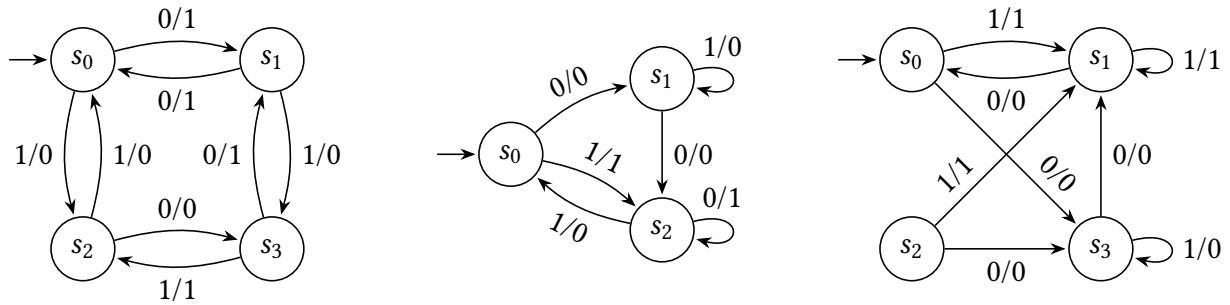
1. For each given regular expression $P$, construct a DFA (Deterministic Finite Automaton), and find the number of accepted word of length at most 5, i.e. the size of the set $\mathcal{L}' = \{w \in \mathcal{L}(P) \mid |w| \leq 5\}$. For "any" (`.`) and "negative" (`[^.]`) matches, assume that the alphabet is $\Sigma = \{\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\}$.
   (a) $P_1 = \mathtt{ab*}$
   (b) $P_2 = \mathtt{a+b?c}$
   (c) $P_3 = \mathtt{[^cd]+c\{3\}}$
   (d) $P_4 = \mathtt{[^a](.|ddd)?}$
   (e) $P_5 = \mathtt{d(a|bc)*}$
   (f) $P_6 = \mathtt{((a|ab)[cd])\{2\}}$

2. Describe the set of strings defined by each of these sets of productions in EBNF[↗] (extended Backus-Naur form).
   (a) $\langle string\rangle$ ::= $\langle L\rangle$+ $\langle D\rangle$? $\langle L\rangle$+
   $\langle L\rangle$ ::= a | b | c
   $\langle D\rangle$ ::= 0 | 1
   (b) $\langle string\rangle$ ::= $\langle sign\rangle$? $\langle N\rangle$
   $\langle sign\rangle$ ::= '+' | '−'
   $\langle N\rangle$ ::= $\langle D\rangle$ ($\langle D\rangle$ | 0)*
   $\langle D\rangle$ ::= 1 | ... | 9
   (c) $\langle string\rangle$ ::= $\langle L\rangle$* ($\langle D\rangle$+)? $\langle L\rangle$*
   $\langle L\rangle$ ::= x | y
   $\langle D\rangle$ ::= 0 | 1
   (d) $\langle string\rangle$ ::= $\langle C\rangle$ $\langle R\rangle$*
   $\langle C\rangle$ ::= a | ... | z | A | ... | Z
   $\langle D\rangle$ ::= 0 | ... | 9
   $\langle R\rangle$ ::= $\langle C\rangle$ | $\langle D\rangle$ | '_'

3. Let $\mathcal{G} = \langle V, T, S, P\rangle$ be the phrase-structure grammar with vocabulary $V = \{\mathtt{A}, \mathtt{S}\}$, terminal symbols $T = \{0, 1\}$, start symbol $S = \mathtt{S}$, and set of productions $P$: $\mathtt{S} \to \mathtt{1S}$, $\mathtt{S} \to \mathtt{00A}$, $\mathtt{A} \to \mathtt{0A}$, $\mathtt{A} \to \mathtt{0}$.
   (a) Show that 111000 belongs to the language generated by $\mathcal{G}$.
   (b) Show that 11001 does not belong to the language generated by $\mathcal{G}$.
   (c) What is the language generated by $\mathcal{G}$?

4. Find the output generated from the input string 01110 for each of the following Mealy machines.



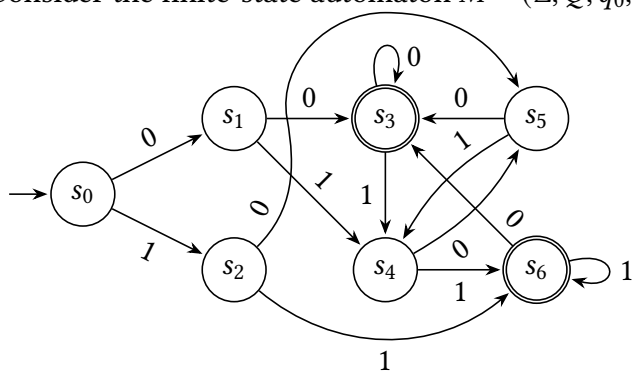

5. Construct a Moore machine for each of the following descriptions.
   (a) Determine the residue modulo 3 of the input treated as a binary number. For example, for input $\varepsilon$ (which corresponds to "value" 0) the residue is 0; 101 (5 in decimal) has residue 2; and 1010 (value 10) has residue 1.
   (b) Output the residue modulo 5 of the input from $\{0, 1, 2\}^*$ treated as a ternary (base 3) number.
   (c) Output $A$ if the binary input ends with 101; output $B$ if it ends with 110; otherwise output $C$.

6. Show that regular languages are *closed* under the following operations.
   (a) Union, that is, if $L_1$ and $L_2$ are regular languages, then $L_1 \cup L_2$ is also regular.
   (b) Concatenation, that is, if $L_1$ and $L_2$ are regular languages, then $L_1 \cdot L_2$ is also regular.
   (c) Kleene star, that is, if $L$ is a regular language, then $L^*$ is also regular.
   (d) Complement, that is, if $L$ is a regular language, then $\overline{L} = \Sigma^* - L$ is also regular.
   (e) Intersection, that is, if $L_1$ and $L_2$ are regular languages, then $L_1 \cap L_2$ is also regular.

7. Determine whether the following languages are regular or not. For non-regular languages, use Pumping lemma to prove that they are not regular. For each regular language, provide a regular expression and construct an $\varepsilon$-NFA.
   (a) $L_1 = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is odd}\}$
   (b) $L_2 = \{0^n 1^n \mid n \in \mathbb{N}\}$
   (c) $L_3 = \{w \in \{0,1\}^* \mid w \text{ contains an even number of 1s}\}$
   (d) $L_4 = \{1^{n^2} \mid n \in \mathbb{N}\}$

8. Consider a finite-state automaton $M = (\Sigma, Q, q_0, F, \delta)$ and a non-negative integer $k$. Let $R_k$ be the relation on the set of states of $M$ such that $s\ R_k\ t$ if and only if for every input string $w \in \Sigma^*$ with $|w| \leq k$, $\delta(s, w)$ and $\delta(t, w)$ are both final states or both not final states. Furthermore, let $R^*$ be the relation on the set of states of $M$ such that $s\ R^*\ t$ if and only if for every input string $w \in \Sigma^*$, regardless of length, $\delta(s, w)$ and $\delta(t, w)$ are both final states or both not final states.
   (a) Show that for every nonnegative integer $k$, $R_k$ is an equivalence relation on $S$.
   Two states $s$ and $t$ are called $k$-equivalent if $s\ R_k\ t$.
   (b) Show that $R^*$ is an equivalence relation on $S$.
   Two states $s$ and $t$ are called $*$-equivalent if $s\ R^*\ t$.
   (c) Show that if two states $s$ and $t$ are $k$-equivalent ($k > 0$), then they are also $(k-1)$-equivalent.
   (d) Show that the equivalence classes of $R_k$ are a *refinement* of the equivalence classes of $R_{k-1}$.
   (e) Show that if two states $s$ and $t$ are $k$-equivalent for every non-negative integer $k$, then they are $*$-equivalent.
   (f) Show that all states in a given $R^*$-equivalence class are final or all are not final.
   (g) Show that if two states $s$ and $t$ are $*$-equivalent, then $\delta(s, a)$ and $\delta(t, a)$ are also $*$-equivalent for all $a \in \Sigma$.

9. Consider the finite-state automaton $M = (\Sigma, Q, q_0, F, \delta)$ depicted below.




   (a) Find the $k$-equivalence classes of $M$ for $k = 0, 1, 2, 3$.
   (b) Find the $*$-equivalence classes of $M$.
   (c) Construct the quotient automaton $\overline{M}$ of $M$.
   ▹ The quotient automaton $\overline{M}$ of the deterministic finite-state automaton $M = (\Sigma, S, s_0, F, \delta)$ is the finite state automaton $\overline{M} = (\Sigma, \overline{S}, [s_0]_{R^*}, \overline{F}, \overline{\delta})$, where the set of states $\overline{S}$ is the set of $R^*$-equivalence classes of $S$; the transition function $\overline{\delta}$ is defined by $\overline{\delta}([s]_{R^*}, a) = [\delta(s, a)]_{R^*}$ for all states $[s]_{R^*}$ of $\overline{M}$ and input symbols $a \in \Sigma$; and $\overline{F}$ is the set consiting of $R^*$-equivalence classes of final states of $M$.

10. Solve the following regex crosswords[1]. Fill each cell with a single ASCII character (an uppercase letter, a digit, a punctuation mark, or a space). Each row/column, when read left to right or top to bottom must match the regular expression(s) given for that row/column.

| | [^SPEAK]+ | EP\|IP\|EF |
|---|---|---|
| HE\|LL\|O+ | | |
| [PLEASE]+ | | |

| | (FI\|A)+ | (YE\|OT)K | (.)[IF]+ | [NODE]+ | (FY\|F\|RG)+ |
|---|---|---|---|---|---|
| (Y\|F)(.)\2[DAF]\1 | | | | | |
| (U\|O\|I)*T[FRO]+ | | | | | |
| [KANE]*[GIN]* | | | | | |

| | [^NRU](NO\|ON) | (D\|FU\|UF)+ | (FO\|A\|R)* | (N\|A)* |
|---|---|---|---|---|
| [RUNT]* | | | | |
| O.*[HAT] | | | | |
| (.)*DO\1 | | | | |

| | [^MCI]+ | .A | (TM\|BF) | |
|---|---|---|---|---|
| (CAT\|A-T)+ | | | | [^KI\sP]+ |
| [MA\-\sE]+ | | | | (M\|APS\|EA)* |
| | [AI][E\s] | [A\-Z]+ | [\sT\-M]+ | |

| | (.)\1(.)\2 | [C\sOU]+ | [^PU\sH]+ | |
|---|---|---|---|---|
| .[LUH]+ | | | | .*L+ |
| (P\|K)[^U]+ | | | | [PUF\s]* |
| .*C+[TIF] | | | | [TIC]* |
| (NO\|ONE\|ION)* | | | | [NOI\sE]+ |
| | [PIF]+ | .*[OWE]* | (TN\|LF\|TF)* | |

| | [^P]I(IT\|ME) | .*E.* | (EM\|FE)(IT\|IP) | (TS\|PE\|KE)* |
|---|---|---|---|---|
| (EP\|ST)* | | | | |
| T[A-Z]* | | | | |
| .M.T | | | | |
| .*P.[S-X]+ | | | | |

[1] Credits: https://regexcrossword.com